

# SOCIAL MEDIA SENTIMENT ANALYSIS.

## Technical Report.

Submitted By,

NAME: Divya Bharathi Ayothi.

Himanshi Gupta.

DEPARTMENT: Electronics and Communication Engineering.

PROJECT TITLE: Social Media Sentiment Analysis.

GROUP NO: 80.

## 1. INTRODUCTION.

Social Media Sentiment Analysis is a web application developed using Python and Flask that analyses the sentiment of user-provided text. The application classifies text into positive, negative, or neutral sentiment. It also maintains a history of analyzed texts through REST APIs. The project demonstrates backend development, API creation, testing, and GitHub version control.

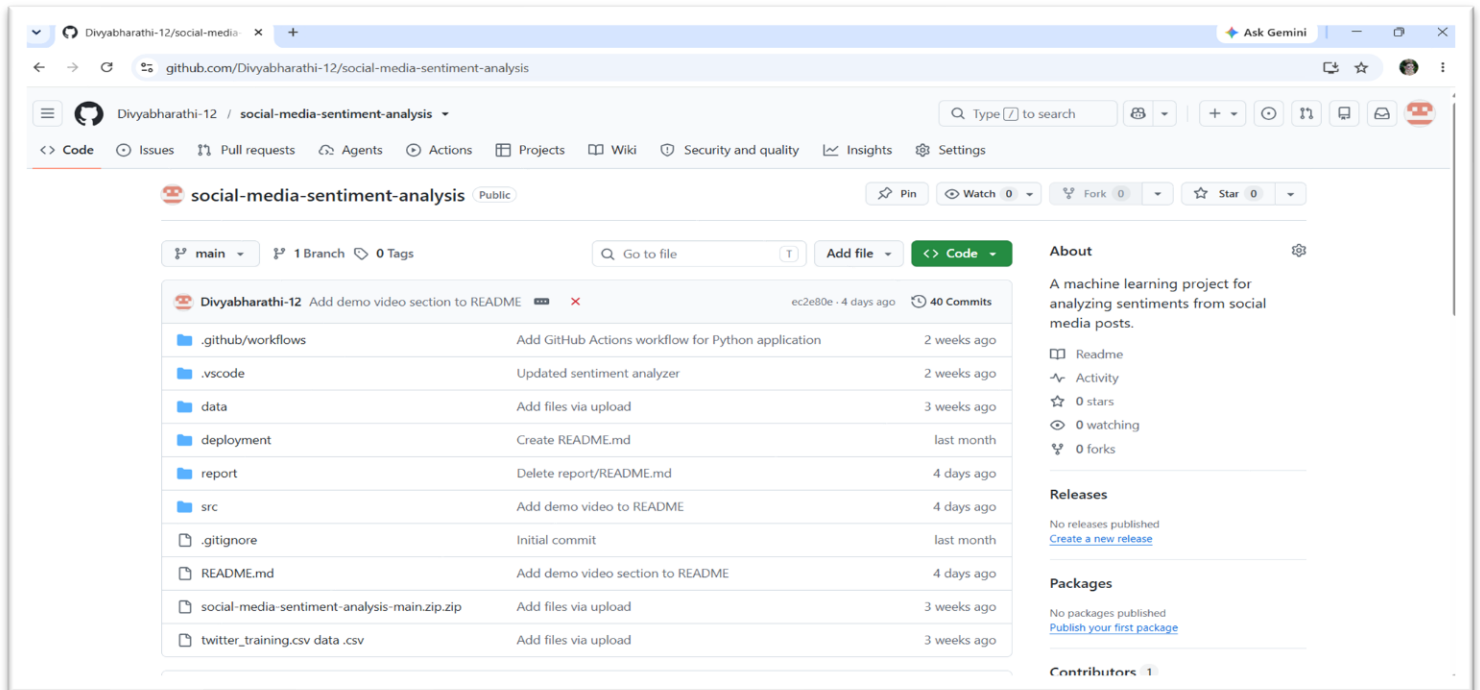
## 2. OBJECTIVES.

- Develop a sentiment analysis web application.
- Create REST APIs using Flask.
- Test APIs using Postman.
- Maintain conversation history.
- Upload the project to GitHub for version control and documentation.

## 3. Tools and Technologies Used:

| Technology            | Purpose                |
|-----------------------|------------------------|
| Python                | Backend Programming    |
| Flask                 | Web Framework          |
| HTML, CSS, JavaScript | Frontend               |
| Postman               | API Testing            |
| GitHub                | Source Code Repository |

## 4. Project Structure.



GitHub Repository Structure.

The project is organized into the following folders:

- **src/**– Contains the source code (Flask backend, HTML, CSS, and JavaScript files).
- **data/**– Contains the dataset used for sentiment analysis.
- **deployment/**– Contains deployment-related files and configuration.
- **report/** – Contains project documentation and reports.
- **README.md** – Provides project description, setup instructions, and usage details.

## 5. Flask Server Execution.

The Flask application was executed successfully on the local server.

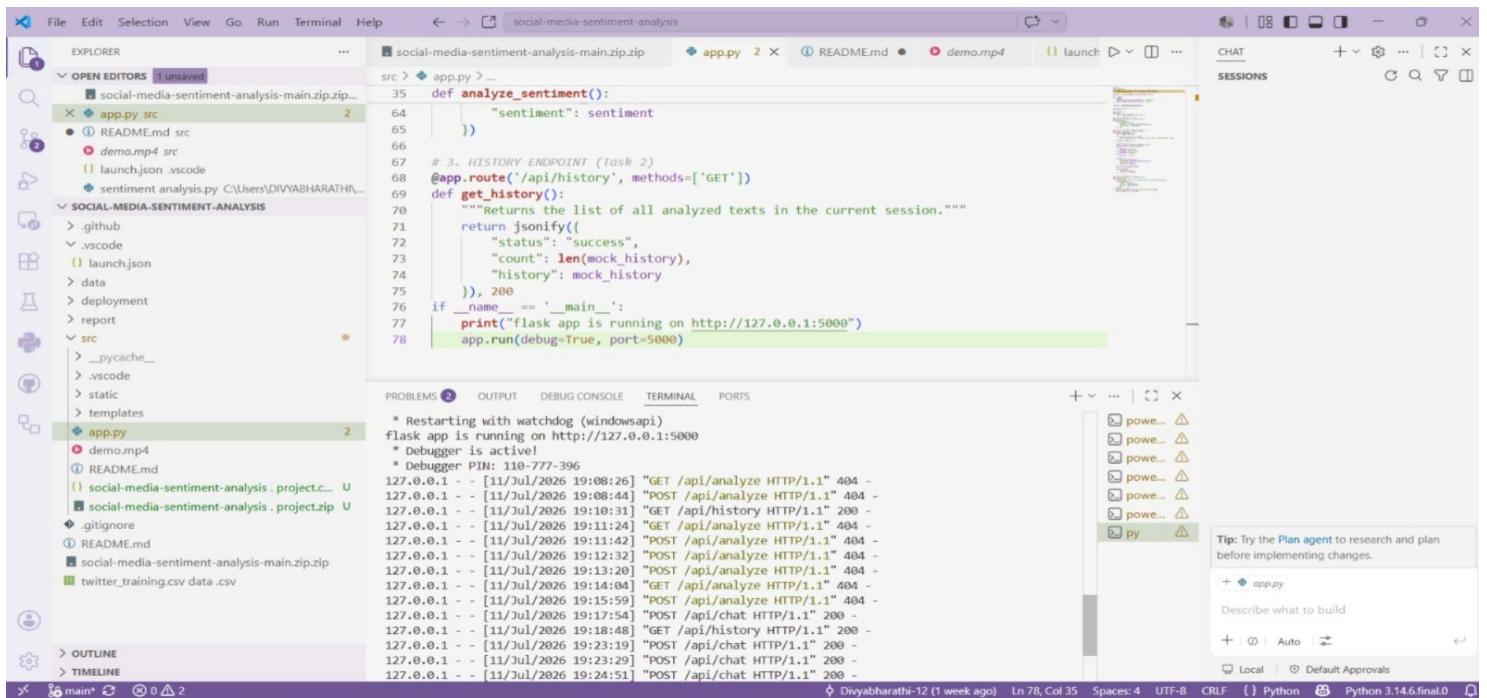
Server Address:

<http://127.0.0.1:5000>

The terminal displayed:

Flask app is running on  
<http://127.0.0.1:5000>

Debugger is active!



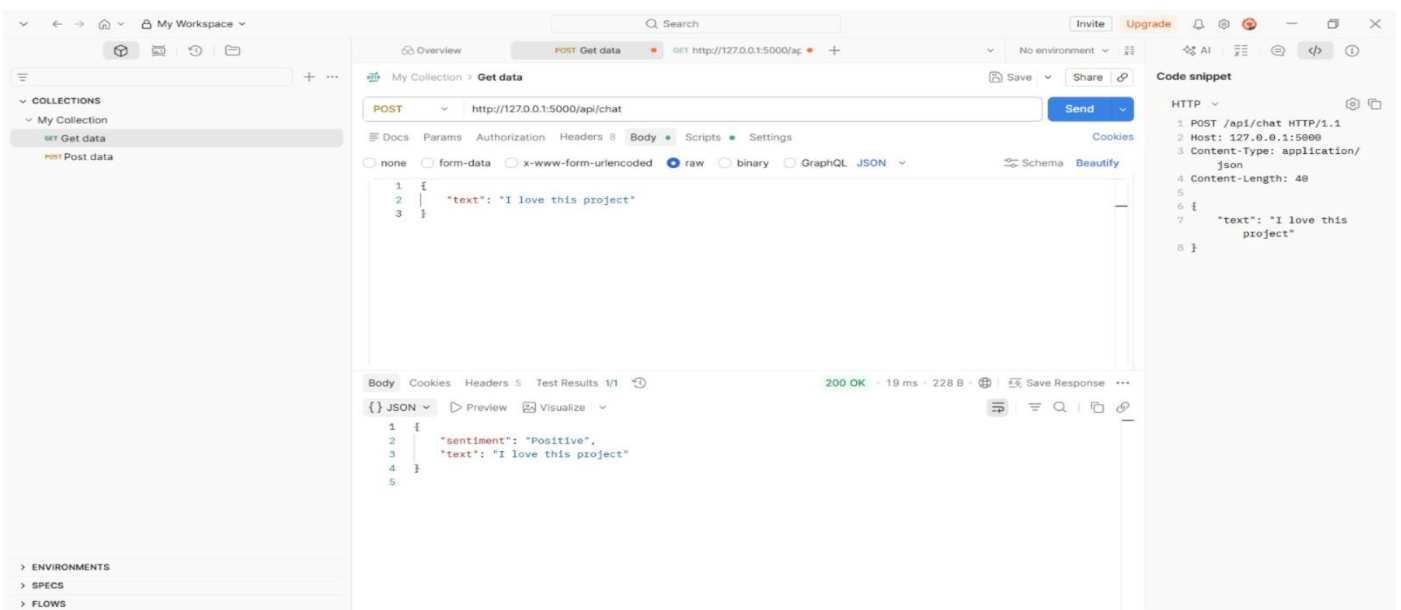
## FLASK SERVER.

### 5. API TESTING.

#### API 1 – Analyze Sentiment

Method: POST

Endpoint: /api/chat

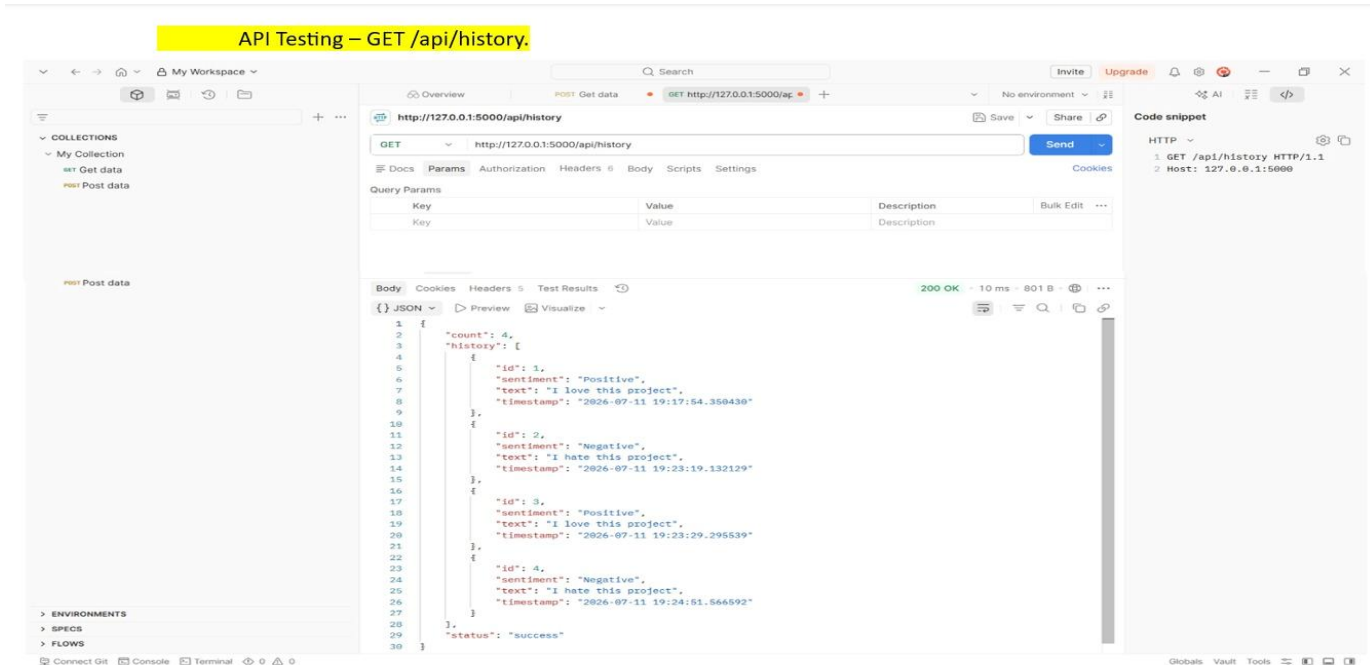


## POST API Response in POSTMAN

## API 2 – Conversation History.

Method: GET

Endpoint: /api/history



## 6. Results

- Flask server executed successfully.
- POST API successfully analyzed sentiment.
- GET API successfully retrieved conversation history.
- APIs returned JSON responses with 200 OK status.
- Project uploaded successfully to GitHub.

## 7. CONCLUSION

The Social Media Sentiment Analysis project was successfully implemented using Python and Flask. REST APIs were created and tested using Postman. The application correctly analyzed user's text and maintained the conversation history. The project was documented and uploaded to GitHub successfully.

GitHub Repository Link.

<https://github.com/Divyabharathi-12/social-media-sentiment-analysis>